AI in Power Plant

From Data to Decisions

An internal faculty programme on Artificial Intelligence for plant operation and maintenance — hands-on, on your own phone

MEASURE · EXPECT · COMPARE · DECIDE

PRESENTER

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- *Lean AI*
- *Learn the English that AI Understands*
- *AI for Busy Parents*

16 years
in the power sector

Koradi Thermal Power Station · 3×660 MW

One-day programme · 10:30–17:00 · six hands-on labs on participants' own phones · all artefacts live at pariharshyamu.github.io/simulator

Two numbers to hold all day

One is a reading nobody can judge alone. The other is what June cost us.

ID FAN A · DE BEARING · ONE READING

74.2 °C

On its own it means nothing. At 45% load on a January night it is alarming; at full load on a May afternoon it is ordinary. No fixed threshold can tell the difference — context can.

KORADI 8–10 · JUNE 2026 GROSS HEAT-RATE GAP

₹22.87 crore

176 kcal/kWh against the norm, in one month, from our own energy bill. Finding where it physically lives is a data problem — the same kind of problem as the bearing.

How does the first kind of number learn to protect the second?

The day, hour by hour

Seven blocks, six hands-on labs, and every block ends on a decision

1	From plant to data	10:30–11:05
2	Neural networks, opened up · Theatre lab	11:05–12:05
3	Data synthesis · Model Builder lab	12:05–13:00
·	Lunch — lab URLs stay on screen	13:00–13:45
4	Predictive maintenance · PdM simulator	13:45–14:50
·	Tea	14:50–15:05
5	Efficiency & optimisation · Simulation Lab	15:05–16:10
6	Computer vision · SafetyNet live	16:10–16:40
7	From data to decisions · pilots	16:40–17:00

6

hands-on labs — five on your own phone, one live model demo

Breaks

Morning tea	11:10 – 11:25
Lunch	13:00 – 14:30
Afternoon tea	15:45 – 15:55

Open the toolkit — two minutes, now

Everything used today lives at one address and runs on your phone

SETUP pariharshyamu.github.io/simulator

2 minutes at your laptop

Do this

1. Scan the QR code or type the address into any phone browser
2. Open the landing page and keep the tab for the whole day
3. Tap 'Predictive Maintenance Simulator' once so it caches
4. Pair up if a phone is short — every lab works in pairs

Watch for

- A landing page with five artefacts, all free, all offline-capable
- No login and no upload — nothing you do today leaves the phone

Every model in this course trains and runs on the device in your pocket. That fact is itself the first lesson: none of this needs a data centre.

The asset you already own

The historian has been quietly building our dataset for years

ONE MACHINE, ONE YEAR, ONE-MINUTE SCAN

21 million readings

About 40 useful tags on a single fan or pump, sampled every minute, is 21 million numbers a year — already in the historian, already paid for.

ADDITIONAL HARDWARE NEEDED TO START

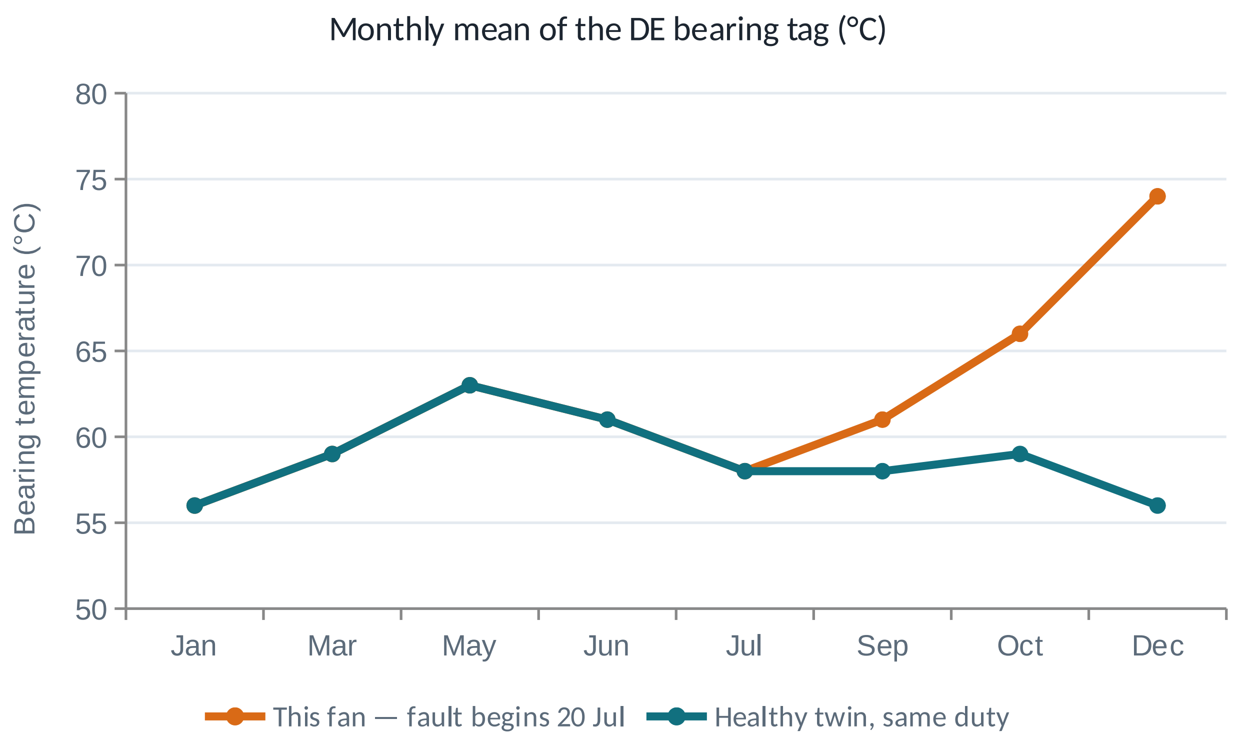
₹0

Every lab today runs on tags a 660 MW unit already historises. The first AI project at this station is an export, not a purchase order.

The historian is the project. The algorithm is the cheap part.

A year of one bearing, raw

Synthetic data, real physics — monthly means of the tag as the historian sees it



Both fans ride the same seasons — the swing is bigger than an early fault

Until September the faulted fan is INSIDE the healthy fan's normal year

A fixed 90 °C alarm sleeps through all of this, then fires near the end

The whole morning is about seeing the gap between these two lines early

For two months the faulted fan looks like a warm summer. That gap is the whole subject.

The one idea of the day

Everything called 'AI' in this course is one subtraction, dressed four ways

1 Measure

What the sensor reads now: 74.2 °C on the DE bearing.

2 Expect

What history says the reading SHOULD be for this load, ambient and cooling-water temperature: 71.9 °C.

3 Subtract

Residual = actual − expected = +2.3 °C. The weather and the duty have been removed; what is left is the machine.

4 Decide

A residual that stays outside its band for days is evidence. Evidence goes to a person, who decides.

Expected − actual = the residual. Flat residual = healthy machine, whatever the weather.

Block 1 close — the data was already there

What we now hold, before any algorithm has been mentioned

The asset

21 million readings per machine per year, already in the historian, already paid for.

The problem

Raw tags hide faults inside load and season. Fixed thresholds are structurally late.

The idea

Expected minus actual. Remove the duty and the weather; what remains is the machine.

The decision

Which of OUR tags could feed a model tomorrow — and what data hygiene blocks them today?

Next: the machine that computes 'expected' — opened up, in 3-D, on your phone.

Six algorithms, one reading

The Algorithm Anatomy Theatre — the same 74.2 °C fed through six different machines

M1 · Nearest neighbours

The residual made geometric: find the 12 most similar past hours, average them. Homework.

M2 · Principal components

The plane the healthy data lives on, and the distance off it. Homework.

M3 · Gradient descent

How every model is actually fitted — a ball rolling downhill. LIVE at 11:15.

M4 · A neural network

163 dials, opened up, trained in your browser. LIVE at 11:30.

M5 · Retrieval

How a question finds the right page — returns in Block 7.

M6 · k-means

The operating regimes nobody labelled. Homework.

Different machines, same conclusion. The algorithm is a means; the residual is the finding.

Three ways machines learn

Ninety seconds of vocabulary, in plant language, then never again

1. **Supervised**
   Learn a mapping from examples with answers: given load, ambient and CW temperature, predict what this bearing should read.

2. **Unsupervised**
   Find structure without answers: cluster a year of operation into regimes — start-up, low load, summer full load — nobody labelled.

3. **Reinforcement**
   Learn actions by trial and reward. Powerful, and not for live plant — nothing today, and nothing this year, should use it on a unit.

4. **What today uses**
   Almost everything in this course is supervised learning of an expected value, plus honest subtraction. Keep the vocabulary that small.

The simplest model that works

k-nearest neighbours: the residual, computed by memory instead of mathematics

1 Take now

This hour's load, ambient and cooling-water temperature — the operating point.

2 Search history

Find the 12 most similar healthy hours in the year. Similarity is just distance.

3 Average them

What the bearing read on those 12 hours, averaged, is the expected value.

4 Subtract

Actual minus expected — the residual again. A lookup table beat the mystique.

Most commercial monitoring systems are this slide, well packaged. Keep that as your baseline.

Lab — Theatre Module 3 · Gradient descent

Twelve minutes: fit a model by rolling a ball downhill, and break it yourself

THEATRE M3

Gradient descent — how a model is actually fitted

12 minutes at your laptop

Do this

1. Open the Theatre from the landing page; pick Module 3
2. Play the acts: the terrain, the ball, the step size
3. Drag the learning rate UP until training visibly diverges
4. Find the largest step that still converges — that tension is real

Watch for

- The loss falling fast, then flattening — the shape of all training
- Too large a step overshooting the valley and climbing the far wall
- The same loss curve you will meet again in the Model Builder

Training is not mystical: it is descent on an error surface. Once you have personally made it diverge, no vendor can hand-wave 'the model learns' past you.

What 'training' means, in one line

The ball, written as algebra — the only equation the day insists on

THE RULE

$$w \leftarrow w - \eta \cdot \partial L / \partial w$$

One dial at a time: move each weight a small step against its own contribution to the error. η is the step size you just tuned by hand in Module 3.

IN PLANT WORDS

Blame, shared out

The error at the output is divided among the dials by the chain rule; each dial moves against its share. Repeat over every hour of history until the error stops falling.

You will see this exact line again after lunch, running live with your own model's numbers.

Lab — Theatre Module 4 · A neural network, opened up

Fifteen minutes inside 163 dials, trained live in your browser

THEATRE M4

A neural network, opened up

15 minutes at your laptop

Do this

1. Open Module 4; run the six acts in order
2. Rotate the network — the weights are a solid volume, not a diagram
3. TAP any neuron: its weights, its sum, its output appear with live numbers
4. Press 'Train from scratch' and watch noise become structure
5. Find a neuron whose output is pinned near ±1 — remember it

Watch for

- 146 weights + 17 biases = 163 dials, each one inspectable
- One neuron's full arithmetic: multiply, add, squash — nothing else
- The loss curve falling exactly like Module 3's ball

Nobody who has tapped a neuron and read its multiplication ever again believes a neural network is magic. That inoculation is worth more than any definition.

What you just tapped

One neuron of the network, exactly as the panel showed it

Theatre M4 · hidden neuron 1 · day 96, 14:00

● Unit load × w	0.93 × +0.41 = +0.38
● Ambient × w	0.55 × +0.29 = +0.16
● Motor current × w	0.95 × +0.29 = +0.28
● ... five more inputs	(each: reading × its own weight)
● + bias	−0.34
● z = sum → a = tanh(z)	z = +1.52 → a = +0.91

Multiply, add, squash. A 'layer' is seventeen of these. That is the whole mystery.

The backward pass, on the same hour

Where the weights come from — the chain rule, sharing out the blame

1 Error at the output

$\delta = 2(\hat{y} - y)$: how wrong, and in which direction, on this hour.

2 Share the blame

The chain rule passes δ backwards; each neuron receives exactly its share.

3 Gradient per dial

$\partial L/\partial w = \delta \times \text{input}$. A dial fed by a big input carries more of the blame.

4 Update

$w \leftarrow w - \eta \cdot \partial L/\partial w$ — Module 3's line, applied 163 times, once per dial.

$\tanh' = 1 - a^2$. A neuron pinned at ±1 has stopped learning — you found one yourself.

Why the bend matters

Remove every activation and the whole stack collapses

WITHOUT ACTIVATIONS

One straight line

Ten linear layers multiply out to a single linear regression. Depth adds nothing — the stack collapses.

WITH A BEND AT EACH NEURON

Any curve

Enough bent units approximate any well-behaved relationship — including the cooler knee you will meet after this block.

Hold that thought: Block 3 has a real knee, and a straight line will lie about it.

The first sanity check on any model

Sensitivity: what the network actually leans on, at the current operating point

Input	Prediction moves by	Physical route to bearing temp	Verdict
Unit load	largest	direct — duty heats the bearing	expected
Ambient	second	direct — cooling air and CW follow it	expected
CW inlet temp	material	direct — through the oil cooler	expected
Hour of day	near zero	none — it only correlates with the others	correctly ignored

If the biggest sensitivity has no physical route to the target, the model learned a coincidence.

What a neural network is — and is not

The honest datasheet, before anyone gets romantic

It is	It is not
A flexible curve-fitter for real non-linearities	A model of the machine — it contains no physics
Inspectable — every weight is a number you can print	'Explainable' the way a manager means the word
Trainable in seconds on a phone, at this size	Reliable outside the range of data it saw
Retrainable after every overhaul	Immune to contaminated training data
Right for combustion, mills, steam temperature	Needed for most single-machine monitoring
One tool on the shelf	**The point. The residual is the point**

Choose the smallest model that works. That is engineering, not fashion.

The regimes nobody labelled

Module 6 in one slide — unsupervised learning as an operating-mode detector

LABELS THE ALGORITHM WAS GIVEN

0

k-means sees only the operating data — load, ambient, flows. Nobody tells it what a regime is.

REGIMES IT FINDS IN OUR SYNTHETIC YEAR

4

Start-up, low-load nights, monsoon operation, summer full load — the modes every operator already knows by feel.

One model per regime beats one model for everything. Module 6 shows the machine that finds them.

Block 2 close — no magic in the box

What changed in the last hour

You watched

Every multiplication of a working neural network — and tapped its neurons open.

You broke

Gradient descent, personally, by turning the learning rate up until it diverged.

You learned

Training = descent on an error surface. The bend is why depth helps. Saturation is why it stalls.

Still open

Where does training data come from — and what happens when it is quietly wrong?

Next: data synthesis, contamination — and a network you build yourself.

Why synthetic data at all

A teaching fault needs a birthday — real historians never confess one

A REAL HISTORIAN EXPORT

No answer key

When did the degradation truly begin? Real data never says — so 'we detected it 40 days early' can never be verified against truth.

THE SYNTHETIC YEAR IN TODAY'S LABS

Onset: day 200

Fault injected on a known day, growing by a known law. Every detection claim today is marked against this answer key, in front of you.

Synthetic data is not fake data. It is the only data with an answer key.

How the year is built

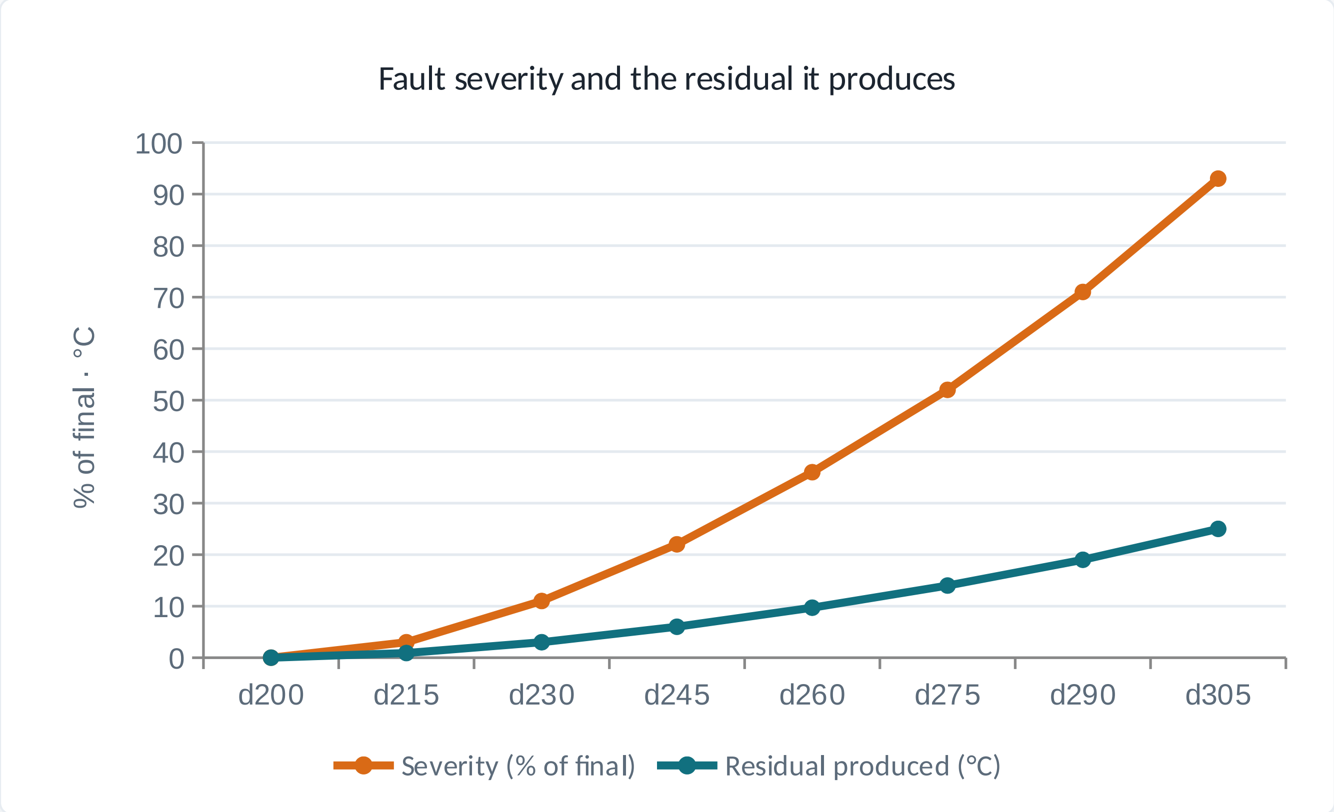
Three context signals, each with the physics that makes winter unlike May

Signal	How it moves	Why it matters
Unit load	Mean-reverts around 80%; nights −13%, Sundays −10%, season ±6%	Duty is the biggest driver of every temperature on the machine
Ambient	Jan 21 °C → May 33.5 °C → monsoon DROP to 25.5 °C — not a sine wave	The monsoon shape is why winter-trained models fail in May
Cooling water	Follows ambient with ~9 days lag — a basin has thermal inertia	The knee (next slide) hangs off this signal
Noise	Gaussian, per sensor, scaled to each instrument	So nothing is suspiciously smooth

One sine wave cannot make an Indian year. Neither can a model that never saw one.

How a fault is injected

Severity grows by a power law — and the first third produces almost nothing



Onset day 200 · 110 days to failure · shape 1.7 — the fan case's actual law

30 days into the fault: 11% severity. Half-life: barely a third

Degradation is convex — the early fault is genuinely quiet

Days of warning are therefore cheap early and expensive late

The first third of a fault produces almost no signal. Remember this at 14:30.

The knee — one line of physics

Why this dataset must be a year long, in a single equation

BELOW 27.5 °C COOLING WATER

Linear

Oil outlet tracks ambient almost perfectly. A straight-line model fitted here looks excellent — on winter.

ABOVE 27.5 °C

$+1.35\cdot(CW-27.5)^{1.45}$

The cooler runs out of approach and oil temperature climbs far faster than the water. Winter models never learned this term exists.

Fit winter, extrapolate May, alarm all summer. You will do exactly this, on purpose, in 20 minutes.

What synthesis cannot teach — real exports are ugly

The pathologies a real historian file brings, and what each does to a model

Pathology	What it looks like	What it does to a model
Status strings	'I/O Timeout', 'Bad', 'Shutdown' inside a number column	Coerced to zero, they teach the model the machine freezes solid
Decimal commas	'61,4' meaning 61.4 — plus semicolon separators	Silently parsed as thousands — every value 10× wrong
Flatlines	A sensor frozen at one value for weeks	The model learns the tag is a constant — and the machine immortal
Compression	Historian dead-banding smoothing real variation away	Noise floor understated; every alert band too tight

The Builder reads exactly this mess — bring one of OUR exports tomorrow and see what it says.

Lab — the Model Builder

Twenty minutes: build and train a real neural network, blocks, no code

BUILDER **Drag-and-drop neural networks on the synthetic year** 20 minutes at your laptop

Do this

1. Open the Model Builder; the default model is ready — press Train
2. Read the verdict panel: advisory day, band, and why it took that long
3. TAP any block — it opens into its own live arithmetic
4. Experiment 2: set the window to days 0–45 and retrain
5. Experiment 3: set the window to days 0–300, then 0–330
6. Experiment 4: tick 'Lube oil cooler outlet' into the inputs

Watch for

- 31 parameters trained in ~1.5 s — on the phone, no cloud
- An advisory on day 255 against a fault that began on day 200
- The maths panel: the same equations as the Theatre, with YOUR numbers

Choosing inputs and windows — not layers — is where models are won and lost, and the only way to believe that is to lose a few yourself in twenty minutes.

Experiment 1 — the honest default

Duty and weather in, bearing temperature out: read the verdict together

`Model Builder · verdict · inputs: fan current, gas flow, ambient`

Residual sigma	2.05 °C — of which sensor noise only 0.89
Alert band	$\pm 4\sigma = \pm 8.2$ °C, held for 72 hours
Fault begins	day 200 (20 Jul) — known, because synthetic
Advisory raised	day 255 (13 Sep) — 55 days later
Severity that day	31% of the way to failure
Episodes this year	1 — a triageable rate

55 days to be sure — and still ~55 days of warning before the machine would have failed.

Experiment 2 — the winter-only window

Train on days 0–45 and the model is confidently wrong all summer

WINTER WINDOW, JUDGED ON WINTER

σ = 0.61 °C

Fits January beautifully — tighter than the honest model. Every bad model looks good on its own training data.

SAME MODEL, MEETING MAY

4× its own band

Runs +2.4 °C wrong on average before any fault exists — 19× worse than the season-spanning window. It never saw the knee.

You just caused seasonal overfitting, on purpose, 90 seconds after being warned about it.

Experiment 3 — the cardinal sin

Let the training window touch the fault, and the model learns the fault is normal

WINDOW 0–300 — TOUCHES THE FAULT

Day 315

60 days of warning lost. The model absorbed the early degradation as 'normal' and raised its own alert band to match.

WINDOW 0–330 — SWALLOWS THE FAULT

Never fires

The model now expects the fault. The residual stays inside the band while the machine walks to failure.

Train on the fault and the model learns the fault is normal. Nothing on the loss curve tells you.

Experiment 4 — buy back 25 days with one tag

The engineer's tag beats the data scientist's architecture

DUTY + WEATHER ONLY

Day 255

σ 2.05 °C, band ±8.2. The model cannot see the cooler, so the knee sits inside its error — as band width.

+ LUBE-OIL COOLER OUTLET

Day 230

σ 0.60 °C, band ±2.4. The mechanism tag explains the knee, the band closes, and 25 days of warning appear.

Input choice beats architecture. The engineer who knows the plant wins.

Where the alert band actually comes from

σ is the model's ignorance, not the instrument's noise

IRREDUCIBLE SENSOR NOISE

0.89 °C

Measured from hour-to-hour differences. No model, however clever, removes this floor.

THE MODEL'S OWN IGNORANCE

1.8 °C

Structure the inputs cannot explain — mostly the knee. This part is yours to remove, and Experiment 4 just removed it.

The alarm band is a property of the model, not the machine. Lunch — leave the Builder open.

Three rules for a training window

Everything Block 3 taught, as procedure — laminate this

1 Span the seasons

A window that has not met May does not know May exists. One full year, minimum, in this climate.

2 End before the fault

The window must end before the period you intend to judge — always, procedurally, no exceptions.

3 Duty in, symptoms out

Inputs are what drives the machine (load, ambient, CW). A symptom tag as input teaches the model to expect the fault.

4 Retrain after overhaul

Maintenance changes 'normal'. A model that remembers the old bearing alarms on the new one.

The training window is a decision. Make it deliberately, and write it down.

From one model to a programme

The eight stages of the PdM simulator — and where programmes actually die

Stages 1–4 · where programmes die

- 01 · Asset & failure mode — decide what you are catching
- 02 · Instrumentation — what is measured, honestly
- 03 · Data acquisition — scan rates, compression, gaps
- 04 · Data quality — flatlines, freezes, timeouts
- Most failed programmes never had a chance past this line

Stages 5–8 · where value appears

- 05 · Feature engineering — normalise, difference, trend
- 06 · Model training — the morning's whole story
- 07 · Validation — alert budgets, sister machines
- 08 · Inference & value — advisory, decision, work order
- The glamour lives here; the leverage lives on the left

Most PdM programmes die in stages 1–4, before any AI exists to blame.

Five machines, five failure modes

The simulator's cases — synthetic datasets, real failure physics

ID fan — DE bearing

Cooler fouls → oil hot → film thins → babbitt wipes. Today's guided case.

HT motor — rotor bar

Broken bar sidebands; the tag that matters is rarely historised.

Coal mill — wear

Grinding capacity walks away slowly; every tag drifts together.

Boiler feed pump

Seal wear and cavitation — two faults that impersonate each other.

Transformer — cooling

A static machine: no vibration to lean on, only heat balance.

All five

One year each, fault day known, traps included — as real programmes have.

Five machines, one method. The fan is guided; the rest are homework with answer keys.

Stage 1 — name the failure mode

A predictive model with no named failure mode is a dashboard

1 Mechanism

Cooling-water side of the oil cooler fouls — slowly, over months.

2 Consequence

Oil outlet temperature rises; the oil film in the journal bearing thins.

3 Damage

Babbitt wears, metal temperature climbs toward the 100 °C trip.

4 Failure

Trip — or a wiped bearing and a forced outage measured in days.

Name the failure mode first. Every link in the chain tells you what to measure.

Stage 2 — what is instrumented, honestly

The fan's eight tags — and the one that is missing

Tag	Fitted?	Role in detection
DE / NDE bearing metal temperature	yes	The signal the fault ends up in — and its twin for triage
Lube-oil cooler outlet temperature	yes	Where the failure actually starts — worth 25 days (Exp. 4)
Bearing housing vibration	yes	Late but unambiguous confirmation
Motor current · gas flow · ambient	yes	Duty and weather — the inputs that define 'expected'
Cooling-water flow through cooler	NO	The mechanism tag. Typically never fitted

1 of 9

the tag closest to the mechanism is the one not instrumented — entirely typical

The P–F curve, with measured days

Where each technique detects the fan fault — from the simulator, not a textbook

Fault begins (P)	Nothing detectable yet — severity zero	day 200
Residual + mechanism tag	Cooler outlet explains the knee; band is tight	~day 230
Residual, duty & weather	The honest default from the morning	day 255
Vibration trend	Late but unambiguous — confirmation, not warning	~day 280
Fixed 90 °C alarm	Structurally late: it waits for absolute damage	~day 290
Trip / failure (F)	100 °C metal temperature, or a wiped bearing	~day 310

Lab — scrub the year

Twelve minutes inside the machine: watch the fault grow in 3-D

PDM · STAGE 3–4

The ID fan case, scrubbed end to end

12 minutes at your laptop

Do this

1. Open the PdM simulator → ID fan case
2. Scrub the day slider slowly from day 0 to day 365
3. Watch the bearing colour, the cut-away, and the live sensor dots
4. Write down the day YOU would have called maintenance
5. Try stage 4: inject data-quality faults and watch the tags lie

Watch for

- A machine that degrades visibly — but only late in the fault
- Sensor values that look ordinary through most of the degradation
- Your own called day, somewhere between 230 and 330 like everyone's

The spread of called days across the room is the argument for residual monitoring: humans eyeball trends inconsistently, especially before the convex fault has produced its signal.

Stage 4 — the data-quality gates

Five ways the data lies before any model sees it

Flatline	A frozen sensor reads one value for weeks. A model trained on it learns the tag is a constant — and misses everything.
Gaps & timeouts	'I/O Timeout' and missing hours, silently zero-filled, teach the model the machine periodically freezes solid.
Compression	Historian dead-banding smooths real variation away; the noise floor is understated and every band is too tight.
Calibration drift	A sensor drifting 2 °C looks exactly like a machine drifting 2 °C — until the twin tag disagrees.
Clock skew	Tags stamped by different systems minutes apart smear every relationship the model must learn.

Never train on: unprofiled tags · zero-filled gaps · a window containing the fault · another unit's data unchecked

Stage 5 — features are engineering

What to hand the model, and the traps that look helpful

Feed the model

- Load- and ambient-normalised temperatures
- DE − NDE difference — common-mode drift cancels
- Oil ΔT across the cooler — the mechanism, directly
- 7-day rolling mean and slope — shift noise suppressed
- Duty tags: current, flow, ambient — 'expected' lives here

Traps that look helpful

- Hour of day — correlates with everything, means nothing
- Day number — encodes the answer outright; leakage
- Any symptom tag as an input — teaches the model the fault
- The twin bearing as input — model copies it and learns nothing
- Kitchen-sink features — more inputs, more coincidences

DE − NDE is the 30-second triage: a sensor fault moves alone, a machine fault moves both.

Stage 6 — model families, honestly

Three ways to compute 'expected' — all of them the same idea

k-NN similarity

No training, fully explainable, robust. What most commercial CM systems really run under the paint.

Regression / small network

The morning's entire story. Interpretable at 31–163 parameters; retrain in seconds.

Autoencoder

Compress and reconstruct; the reconstruction error is the residual. PCA with bends — for many tags at once.

All of them

Produce expected − actual. Window, inputs and validation decide success; the family mostly decides the brochure.

Every family computes the same subtraction. Spend your judgement on windows and inputs, not brand names.

Stage 7 — validation that means something

The alert budget: how many pages a year buys how many days of warning

Rule	Episodes / year	Warning on the fan case	Verdict
4σ · 72 h	1	55 days	Adoptable — the default
3σ · 24 h	≈ 4–6	≈ 60 days	Only with a named triage owner
2σ · 24 h	dozens	marginally more	Ignored within a month
4σ · 72 h + mechanism tag	1	80 days	Where Experiment 4 lands

2 / month

a realistic triage budget for one unit's fleet of monitored machines — set it BEFORE tuning σ

The sister-machine check

The negative control every claimed detection must pass

FAN A — THE FAULTED CASE

1 episode

Advisory day 255, sustained, single episode. The detection claim.

FAN B — HEALTHY TWIN, OWN THRESHOLD

0 episodes

Same model, same year, zero alarms. If this number is not zero, the model learned the season — not the fault.

No sister-machine run, no belief. Ask for it in every vendor demo.

Remaining useful life — a band, not a date

The difference between astrology and engineering

WHAT GETS ASKED FOR

"Fails 14 March"

A single confident date. It will be wrong, it will be remembered, and it will cost the programme its credibility.

WHAT ENGINEERING CAN HONESTLY SAY

P10–P90 band

'90% it survives past the outage window; here is what moves the odds.' Planning can act on this — advance, stage spares, derate.

A date is astrology. A band with named drivers is engineering.

Stage 8 — the advisory, as it should arrive

Not an alarm: a claim, with its evidence attached

Unit · ID fan A · advisory · day 255, 14:20

Claim	DE bearing running above expected for duty & weather
Evidence	Residual +8.4 °C sustained 72 h · 1st episode this year
Mechanism	Oil cooler approach shrinking — CW-side fouling suspected
Second witness	Oil outlet ΔT confirms; NDE bearing does NOT follow
Remaining life	P10–P90: 40–75 days at current duty
Recommended	Inspect cooler CW side at next opportunity ≤ 21 days

An advisory is a claim with evidence, addressed to an engineer — not an alarm addressed to nobody.

The availability arithmetic

Scenario numbers from the simulator — the shape is what matters

Path	Outage	Direct + energy cost	Note
Act on day-255 advisory	0.4 day planned	≈ ₹35 lakh	Scheduled into a merit-order dip
Run to failure	≈ 2.5 days forced	≈ ₹4+ crore	Replacement energy at market, overtime, secondary damage
The ratio	—	≈ 10 : 1	Typical for bearing-class failures — scenario, not history

72.5%

our June 2026 availability — every avoided forced outage buys back a piece of this

The vote

It is day 255. The advisory is on your desk. Sign, or wait?

SIGN NOW

≈ ₹35 lakh

0.4-day planned window, cooler inspection and clean. If the mechanism hypothesis is right, the fault is arrested cheaply.

WAIT FOR MORE EVIDENCE

Risk ≈ ₹4+ cr

Severity ~45% and convex. More certainty is coming — but at 10:1 consequence odds and shrinking room to schedule.

Vote now, hands up. This moment is the decision layer — everything today serves it.

Reading our own bill

Three documents, zero estimates — where every number in this block comes from

Energy bill · June 2026

Generation, heat rates, auxiliary consumption, availability — station by station, signed.

Part-I FSA + F10 ECR

What our coal actually cost, per kcal and per kWh, after quality adjustment.

MERC merit order · July 2026

Where every MAHAGENCO station ranks against the market, in ₹/kWh.

The rule

No number without a document. Vendors estimate; bills do not.

Everything in this block is on paper the station already signs.

Koradi 8–10, June 2026, one table

Our station's month, exactly as filed

Metric	Actual	Norm	Gap
Availability / PLF	72.5% / 66.62%	—	964.8 MU sent out gross
Net heat rate	2,442 kcal/kWh	2,230	+212 kcal/kWh
Auxiliary power	6.95%	6.0%	+0.95 pp ≈ 9.2 MU
As-fired GCV / variable cost	3,061 kcal/kg · ₹3.378/kWh	MOD ₹3.284	94 paise above the stack

₹25.61 cr

what the 212 kcal/kWh net heat-rate gap cost in June 2026 alone

The decomposition that changes the conversation

Gross HR = net HR × (1 − aux): split the gap before prescribing the medicine

BOILER–TURBINE (GROSS) GAP

176 kcal · ₹22.87 cr

2,272 gross actual against 2,096 norm. A real thermodynamic shortfall — condenser, heaters, sprays, part-load — worth ₹22.87 crore in June.

AUXILIARY EXCESS

0.95 pp · ₹3.01 cr

6.95% against a 6.0% norm ≈ 9.2 MU of house load. Real, worth chasing — and one-seventh the size of the first number.

Same bill, two different diseases. Point the AI at the ₹22.87 crore one.

Never add those two numbers

The most expensive arithmetic error available in a station review

TEMPTING

25.6 + 3.0 = ₹28.6 cr

Wrong. Net heat rate already contains auxiliary power; this sum counts the same MU twice.

CORRECT

22.87 + 3.01, decomposed

Split via gross HR = net HR × (1 − aux), then the two numbers are independent and may be discussed — separately.

Decompose first. Never add a net-side number to an aux-side number.

Lab — SIM-2 · Heat rate loss attribution

Twelve minutes: find plausible homes for OUR 176 kcal/kWh

SIM-2

Heat rate loss attribution — Koradi 8–10, June 2026

12 minutes at your laptop

Do this

1. Open SIM-2; confirm it is showing the Koradi 8–10 row
2. Move condenser back pressure until you have claimed ~60 kcal
3. Take one HP heater out of service; watch the attribution shift
4. Add spray flow and auxiliary until the 176 kcal gap is 'explained'
5. Note YOUR combination — and compare with your neighbour's

Watch for

- Several different slider combinations that all 'explain' 176 kcal
- The rupee counter converting every kcal to ₹ crore per month
- Why the monthly bill cannot arbitrate between explanations

The gap is real but the bill cannot localise it. Continuous tag-level attribution — this exercise, automated hourly — is the honest AI project this station's numbers are asking for.

Where 176 kcal/kWh physically lives

Candidate homes, sized honestly — controllable and structural

Candidate	Typical share of a gap like ours	Controllable this quarter?
Condenser vacuum · CW system	the classic largest single account	Largely — cleaning, air ingress, CW optimisation
Cycle isolation · HP heaters	material when a heater is out or passing	Yes — find it, fix it, verify the kcal returned
SH/RH sprays · attemperation	moderate but chronic	Partly — control tuning and combustion balance
Part-load operation at 66% PLF	significant and real	No — structural; merit order sets it. Say so honestly

AI localises the controllable share. It does not repeal thermodynamics — and must not promise to.

What we already beat

Two norms this station outperforms — credibility before the gap-chasing

SECONDARY FUEL OIL

0.41 vs 0.50

ml/kWh, actual against norm. Stable combustion support discipline — already better than the standard.

COAL TRANSIT LOSS

0.49% vs 0.80%

Nearly half the normative allowance. The fuel-handling chain is performing.

The bill is not uniformly bad — it is specifically bad. Specific problems get solved.

Auxiliary power is a scheduling problem

SIM-3's equipment models — the arithmetic under the 0.95 excess points

Equipment	Power model (SIM-3)	The scheduling question
Coal mill	340 kW + 2.6 kW per t/h above 30	How many mills for this load and GCV — with margin?
CW pump	1,650 kW each	Two pumps or three, for this season's CW temperature?
ESP	26 kW per energised field	How many fields for compliance at this load and ash?
ID fan	≈ cubic in gas flow	What does every extra mmWC of draught really cost?

Auxiliary excess is mostly a counting problem — and counting problems can be optimised.

Lab — SIM-3 · Auxiliary power optimiser

Ten minutes, scoreboard rules: beat the baseline house load

SIM-3 **Auxiliary power optimiser — 500 MW class** 10 minutes at your laptop

Do this

1. Open SIM-3; take the default load, GCV and season
2. Choose mills, CW pumps and ESP fields; watch kW and feasibility
3. Try one fewer CW pump in a cool-season scenario
4. Push until the simulator flags infeasible — know where the wall is
5. Shout out your best % saving — scoreboard rules

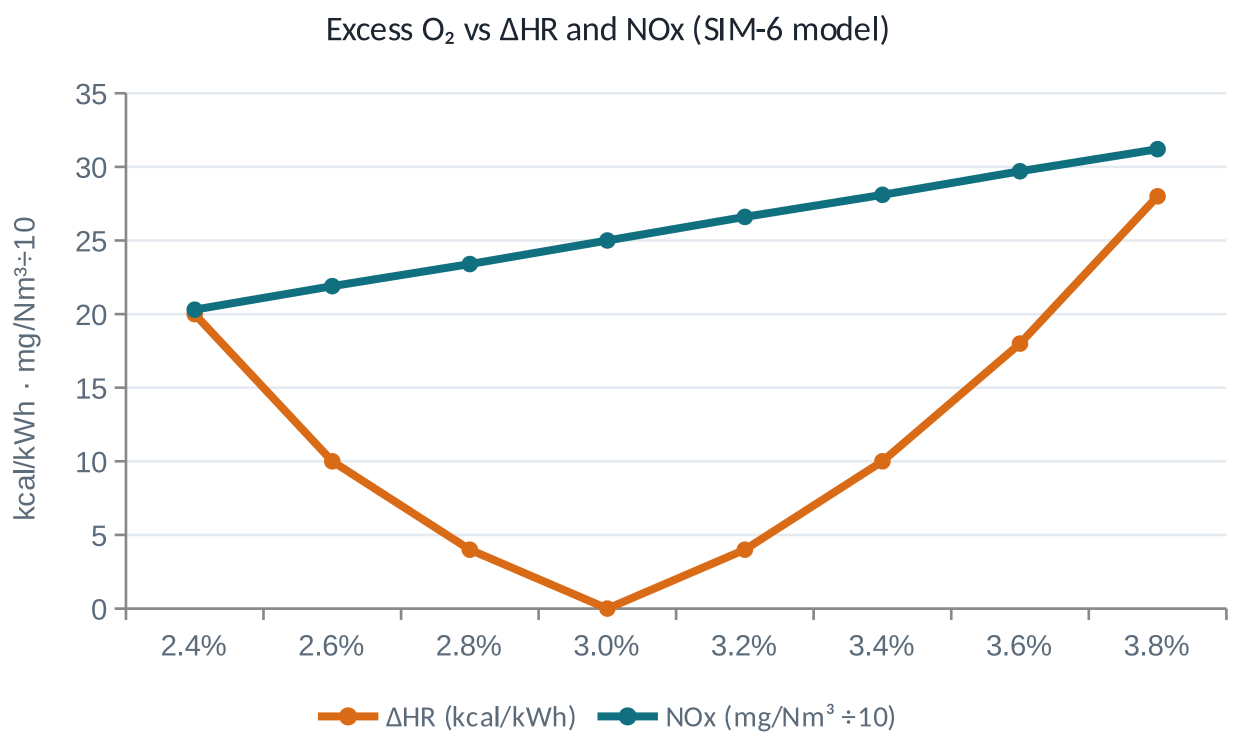
Watch for

- House load responding in real kW to every scheduling choice
- Feasibility flags when milling margin or condenser limits bite
- A recurring finding: one pump too many at part load, off-summer

A 0.95-point auxiliary excess ≈ ₹3 crore a month is largely made of scheduling slack like this — recoverable with discipline and zero capital.

Combustion is a trade-off, not a setting

SIM-6's surfaces: excess O_2 against heat rate and NOx, from the model itself



NOx climbs ~78 mg/Nm³ per point of excess O_2 — SIM-6's own coefficient

Heat rate is U-shaped: dry-gas loss right, CO and unburnt carbon left

Every point on the frontier is defensible; choosing one is policy

OFA, tilt and bias move the frontier itself — that is the optimiser's job

'The optimiser says' is not a complete sentence. Weights are policy; policy is yours.

Lab — SIM-6 · Combustion multi-objective optimiser

Eight minutes on the Pareto frontier

SIM-6

Combustion multi-objective optimiser

8 minutes at your laptop

Do this

1. Open SIM-6; find the three weight sliders
2. Put all weight on heat rate — note NOx and CO
3. Put all weight on NOx — note what heat rate pays
4. Find a blend you would defend in a morning meeting
5. Read the recommended O_2, OFA and tilt for your blend

Watch for

- The operating point sliding along a frontier, not jumping to 'best'
- Every objective improved only by paying another one
- A recommendation with reasons — the shape advisories should take

Multi-objective is the honest formulation of combustion. Single-number 'optimum' claims hide a weight choice someone else made for you.

What the advisory looks like at 14:20

Optimisation, delivered the same way as the bearing advisory

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Unit · combustion advisory · 14:20
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Suggest	O_2 3.4 → 3.1% · OFA +4% · tilt −2°
Expected	−9 kcal/kWh ≈ ₹1.2 cr/quarter at current load
Respecting	NOx ≤ limit − margin · mill margin ≥ 8%
Evidence	Similar operating days, last 90 days, this coal band
Operator	Accept · Decline (reason) — declines retrain the model
Never	Writes to DCS. Advisory only, by design

Same discipline as the bearing: a claim, evidence, a human decision. Never a hand on the controls.

The missing instrument — soft sensors

SIM-4 in one slide: coal quality every minute instead of every shift

LABORATORY GCV

1 per shift

Accurate, authoritative — and hours late. The optimiser spends the whole shift assuming the morning's coal.

SOFT-SENSOR GCV

every minute

Inferred from feeder rate, mill power and heat output. Calibrated against the lab — running between its samples.

Every optimiser today is only as good as the coal number it is fed. SIM-4 is homework.

Adani Udupi — the nearest real deployment

One industry case, read the way an engineer should read it

What it is

AVEVA PI + ML at Udupi: 15-minute efficiency windows, health indices on key pumps and motors.

What it proves

The historian-plus-models architecture runs on Indian coal units in production. Feasibility, settled.

What it does not

No audited ₹ savings published. It cannot size OUR prize — only our bill can (and just did).

What we take

The architecture pattern — and the discipline of 15-minute windows over monthly arguments.

Proof of architecture, not proof of value. Our value number comes from our own bill.

Block 5 close — the card

One candidate, in your handwriting, before tea-less sprint to vision

1 Think

Where, physically, does OUR 176 kcal sit? Your unit, your knowledge.

2 Write

One candidate per card. Specific: equipment, unit, suspicion.

3 Hand in

Cards collected now — anonymous is fine, specific is better.

4 Return

At 16:40 the cards meet the pilot criteria. The best one may become the pilot.

The bill gave the size. Your cards give the addresses.

A camera is a very wide sensor

The modest, true proposition for vision in a plant

AN ENGINEER, ONCE

1 inspection

Expert, contextual, accountable — and finite. Rounds happen once a shift, attention is scarce, fatigue is real.

A CAMERA, ALL DAY

100+ / day

The same simple check — repeated relentlessly, in heat and dust and at 03:00, without boredom. Volume, not brilliance.

Vision is volume, not brilliance. The judgement stays with the engineer.

How a vision model learns

The same story as this morning — pixels in, backprop underneath

1 Pixels in

An image is a grid of numbers — brightness per pixel. Nothing more exotic enters.

2 Convolution

A small filter slides over the grid, reusing its weights — learns 'edge' once, finds it everywhere.

3 Same forward pass

Layers, bends, an output: helmet / no-helmet with a confidence. The Theatre's story, scaled.

4 Same backprop

Blame shared by the chain rule across millions of dials. Same rule, same risks, same windows.

You already learned vision this morning. Only the sensor got wider.

What a vision model needs from you

The dataset is the project — again — plus glass that stays clean

Labelled data	Thousands of annotated images per class, covering YOUR site's conditions — not stock photos of clean workers.
Lighting & angle	A model trained on day shots misses night. Move the camera 2 metres and precision quietly falls.
Lens maintenance	Coal dust on glass is a frozen sensor. Camera cleaning is now an instrument-maintenance route.
Domain shift	New uniforms, new season, new contractor PPE colours — retrain triggers, exactly like after an overhaul.
Honest metrics	Precision AND recall, on your own footage, per class — not a vendor's demo reel.

Never point a first vision project at: face identification · discipline · protection systems · anything that names a worker

Live — SafetyNet

A PPE model trained by an engineer at this station, run in front of the room

SAFETYNET **Per-worker PPE compliance — live** 10 minutes at your laptop

Do this

1. Laptop: python service/gradio_app.py — localhost, offline once weights are cached
2. Image tab: run two or three plant-style photos
3. Show detections AND per-worker verdicts side by side
4. Show one UNKNOWN verdict and explain why it refused to judge
5. Change one policy line (required items) and re-run — no retraining
6. If anything misfires live, teach from the failure

Watch for

- Per-worker verdicts, not 'is there a helmet somewhere in this image'
- An honest UNKNOWN on a small or edge-clipped worker
- Policy changes taking effect instantly, with the model untouched

A working model trained by a station engineer on a free GPU collapses the distance between 'AI programme' and 'us'. The barrier was never hardware or budget.

The question nobody's dataset answers

From 'is there a hardhat in this image?' to 'is THAT worker wearing one?'

1 Detect

10-class YOLO finds people and PPE as independent boxes. The only GPU stage.

2 Attribute

Each PPE box matched to at most one worker: containment × anatomy × scale, as a soft AND.

3 Judge

Site policy decides what the attribution means — including an honest UNKNOWN.

4 Debounce

A violation must persist across frames before it becomes an event. One event, not 400 frames.

mAP measures the detector. The site's question is answered two stages later — measure THOSE.

SafetyNet's honest numbers

What it measures, what it misses, what it refuses to guess

Metric	Value	What it means on the ground
Precision	0.921	When it flags a violation, it is right ~9 times in 10
Recall	0.787	It MISSES ~1 in 5 — a screening aid, not a guarantee
mAP50 / mAP50-95	0.852 / 0.620	Detector quality — necessary, nowhere near sufficient
Abstention	UNKNOWN	<64 px worker, edge-clipped, or ambiguous → refuses to judge

19 MB

the entire trained model — YOLO11s, 120 epochs, free Colab T4, ~90 minutes, zero procurement

Policy out of the model

compliance.yaml — the same argument as the alert-rule block, in pixels

Lives in the model (retrain to change)	Lives in compliance.yaml (edit to change)
What a hardhat looks like	Which items are required in which area
What a person looks like	Minimum worker size before judging
Detection confidence per box	The ambiguity margin that forces UNKNOWN
—	How long a violation persists before an event
—	Whether synthetic (orphan-PPE) workers may alarm
Changes need a GPU and a dataset	**Changes need a text editor and sign-off**

The detector reports; the policy decides. Same principle as the alert rule this morning.

The industrial-relations question

Safety system or disciplinary system? Decided by the FIRST alert

Name-and-shame	The first individually-targeted alert converts a safety tool into a surveillance grievance. Permanently.
False accusation	At 0.921 precision, 1 flag in 13 is wrong. Against a named worker, that is a formal grievance waiting.
Discipline creep	If HR ever asks for the footage, the safety framing is dead — and so is the data source.
Union sequencing	Briefed first, with the miss-rate on the table, unions often support zone-level safety statistics.
Metric theatre	Zone compliance trending up may mean PPE improved — or workers learned the camera positions. Audit on foot.

Never use PPE vision for: identifying individuals · disciplinary evidence · performance appraisal · anything beyond zone statistics

Where plant vision pays first

Ranked honestly — thermal before optical, statistics before individuals

Coal stockpile hot spots	Thermal imaging finds spontaneous combustion early	Proven — quick win
Drone stockpile volumetrics	Photogrammetry replaces surveyors at height	Proven — recommended first
Switchyard / joint hot spots	Scheduled thermal rounds, trended over time	Proven — extends existing practice
Conveyor: hot idlers, tracking	Thermal + edge detection along the run	Emerging — high value in CHP
PPE zone statistics	SafetyNet-class, zone-level only, union-first	Emerging — governance decides
Flame / combustion vision	Camera-based flame analysis feeding tuning	Emerging — value depends on baseline

Start where the camera measures physics. Start where nobody's name is involved.

Two independent witnesses

Vision and tags, corroborating — the day's discipline, unified

THE CAMERA SAYS

Hot idler, bay 7

Thermal spot 18 °C above neighbours, persisting across frames. A claim — from pixels.

THE TAGS SAY

Power ↑ at same tonnage

Drive power residual drifting high for a week on the same belt. The same claim — from the historian.

A vision alert is a claim, like day 255. Corroborate, then act.

The other half of plant knowledge

The historian holds the numbers. The documents hold everything else.

WHAT AN LLM IS

Next-word prediction

Trained on public text. It contains none of our SOPs, limits or revisions — and no signal for when it is guessing.

THE DANGER MODE

Fluent wrongness

A confident, well-phrased answer citing nothing. Near a PTW or an OEM limit, fluency without provenance is a hazard.

The failure mode is not gibberish — it is confidence. Grounding, not size, is the fix.

RAG — the answer must quote our document

Retrieve first, answer second, cite always

1 Chunk & index

Split manuals and SOPs into passages; index by meaning, tagged with document, revision, page.

2 Retrieve

The question finds its passages first — recall@5 of 90% hybrid on the session's measured bench.

3 Answer, grounded

The model writes ONLY from what was retrieved. Nothing retrieved → 'not found', honestly.

4 Cite

Document, revision, page on every claim. Trust check: citation present · doc current · numbers recomputed.

Metadata beat the model: titles and revisions moved recall more than any algorithm choice.

Mini-lab — Pocket RAG, on your phone

Your own PDF, parsed, indexed and answered on the device — nothing leaves it

POCKET RAG

On-device retrieval over your own document

5 minutes at your laptop

Do this

1. Open Pocket RAG from the landing page
2. Load any PDF on your phone — or the sample SOP provided
3. Ask one question you know the answer to
4. Tap the citation and verify the quoted passage
5. Ask one question the document does NOT answer — watch it say so

Watch for

- Parsing, indexing and answering happening entirely on the handset
- A citation on every claim — and an honest miss when nothing matches
- Flight mode still works once the page has loaded

On-device changes the confidentiality conversation completely. Verify-the-citation is the habit that makes RAG safe to use near real procedures.

Why these programmes fail

Eight failure modes — every one demonstrated on your phone today

WHY IT FAILED	WHAT PREVENTS IT
Training window touched the fault	Window ends before the judged period — procedurally, always
Winter window, summer alarms	Span the seasons; retrain after overhaul, every time
Alert spam killed adoption	Fix the alert budget FIRST; spend it with σ and persistence
Symptom tags fed as inputs	Duty in, symptoms out — or the model expects the fault
No named triage owner	Every advisory has a recipient before go-live, or none fire
Model never retrained	Overhauls change 'normal' — retraining is maintenance
Vendor black box	Demand the sensitivity table and the sister-machine run
Ungrounded AI answers	Citation or it does not count — especially near a PTW

None of these is an algorithm failure. All of them are decisions nobody made.

Choosing the first pilot — against your cards

The criteria, the candidates, and ninety days of proof

- Tags already historised, one year clean
- A failure mode this station has actually felt
- Money visible monthly, from the bill
- One named owner for triage
- Provable inside 90 days
- No capex, no procurement cycle
- A sister machine for the negative control

Candidate pilot	Value	Risk	Verdict
Fans + BFPs, one 660 unit	High	Low	START — 90-day proof
176-kcal loss accounting	High	Medium	Second — bigger money, harder data
Aux scheduling advisory	Medium	Low	Fold into the second
GCV soft sensor	Medium	Medium	Third — feeds the optimisers
SafetyNet zone statistics	Medium	Medium	Governance first, then trial
RAG on station SOPs	Medium	Low	Quiet parallel — document hygiene

A 90-day starter plan

Days 1–30

Tag list, one-year export, quality profile, training windows chosen and written down

Pilot owner + C&I

Days 31–60

Models trained; alert budget set at 2/month; sister machines wired as negative controls

Pilot owner

Days 61–90

Live advisories into the morning meeting; every one triaged, closed and costed

Triage owner + HOD

The two pilots, side by side

Different money, different data problems, one shared foundation

Pilot 1 · condition monitoring	Pilot 2 · the 176-kcal hunt
Fans + BFPs of one 660 unit	Condenser, heaters, sprays — continuous attribution
Money: availability protected	Money: ₹22.87 cr/month gap localised
Data: tags we already trust	Data: performance-grade quality needed
Provable in 90 days	Provable in a season
Risk: alert fatigue — budget fixes it	Risk: attribution disputes — measurement fixes them
Proves the discipline	**Spends the discipline on the big number**

Phase 1 proves the discipline. Phase 2 points it at ₹22.87 crore. Same export, same owners.

The safety boundary

What AI may never touch here — policy, not preference

NEVER — Protection systems, interlocks, trip logic, PTW authority, closed-loop control of anything — and identifying individuals on camera

WITH CONTROLS — Operator-facing advisories with named owners; setpoint suggestions a human accepts or declines with a reason; zone-level safety statistics

FREELY — Historian analytics, residual monitoring, loss accounting, document retrieval with citations, training and simulation — everything in this course

Air-gap respected

Everything today ran client-side. Plant systems export data OUT; nothing connects IN.

No cloud dependency

Phone labs cache offline. Pilot models run on-prem hardware the station already owns.

Data leaves by policy

Historian exports follow the same approval as any data leaving today. RAG stays on-device or on-prem.

Model files are files

Version them, hash them, back them up — the 19 MB SafetyNet weight file is config, not infrastructure.

Audit by design

Every advisory logged with evidence and outcome. The JSONL trail is the programme's defence in any review.

The two numbers, reconnected

Where the morning's reading and the bill's crore ended up

THE READING'S JOURNEY

74.2 °C → work order

Sensor → residual → advisory day 255 → the room's vote → a signed decision. You ran every step yourselves.

THE CRORE'S JOURNEY

₹22.87 cr → a pilot

Bill → decomposition → SIM-2 localisation → your cards → a 90-day plan with named owners.

Measure · expect · compare · decide. That is the whole method — for degrees and for crores.

FROM DATA TO DECISIONS

The data is already in our historian. The decision layer is us.

What you can do on Monday morning

1. Send me your unit's five worst actors — machine, failure history, and which tags they carry
2. Run the Model Builder on ONE real historian export from your area — the CSV reader handles our format
3. Name the triage owner for your area: who receives an advisory if one arrives tomorrow morning

The data is already in our historian. The decision layer is us.

Questions — and the cards: which candidate becomes the pilot?